

The following pages contain an artifact sample for **Goal 1b Written Communication** and **Goal 2 Critical Reasoning**.

BUS 480 Strategic Management and Design Thinking

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Educate AI

As artificial intelligence advances at an extraordinary pace, it is rapidly transforming industries and people's lives across the globe, the education world is no exception. Its integration into education presents a significant opportunity, as AI has the potential to personalize learning to meet individual student needs and enhance educators' efficiency and quality of teaching. This raises an important question: How can we integrate AI into the learning process in a way that enhances education without compromising academic integrity or creating overreliance?

Our team proposes EducateAI, an AI-powered learning assistant designed to support both students and educators. This assistant provides guided problem-solving and real-time feedback to promote critical thinking while personalizing education to fit each student's needs. Additionally, it supports educators with lesson planning and grading, allowing them to focus more on engaging with students. EducateAI will seamlessly integrate with Learning Management Systems such as Canvas and Blackboard. It includes built-in privacy protections that comply with the Family Educational Rights and Privacy Act (FERPA) and the General Data Protection Regulation (GDPR), addressing concerns about data ethics and security. Unlike many existing AI tools, EducateAI is designed to balance the use of technology with the need to maintain independent learning and not compromise integrity.

Identifying the Opportunity

The Problem:

In our current education system, there are two extremes in AI usage: complete avoidance or misuse that threatens learning. On one end, some students misuse AI to complete tasks without fully understanding the material, not integrating it effectively into education. On the other end, some students and educators completely avoid using AI, missing the opportunity to enhance learning and teaching. This disconnect between technology and education has left many students unprepared for a job market that increasingly demands more advanced digital skills. At the same time, educators face several challenges including technological distractions in the classroom, privacy and security concerns, and a lack of clear ethical guidelines for AI use. They are also

burdened with heavy workloads that could be avoided through AI integration. Many teachers express concerns about plagiarism, student overreliance on AI, and the absence of effective tools to manage AI in educational settings.

The Opportunity

The AI-in-education market is growing rapidly, with projections indicating a 30% compound annual growth rate from 2023 to 2028. This growth reflects increasing demand for personalized learning experiences, needs that traditional teaching methods can not fully meet. However, existing AI solutions lack ethical safeguards, encourage overreliance, or fail to integrate effectively with the Learning Management Systems used by most educational institutions.

EducateAI is the perfect solution. Our learning assistant fosters independent learning by guiding students through problem-solving rather than simply providing answers. It also supports educators by streamlining tasks such as lesson planning and grading, enabling them to focus more on meaningful student engagement. EducateAI is built with privacy and ethics at its core, offering full compliance with both FERPA and GDPR. Beyond addressing current challenges, EducateAI prepares students for a workforce that increasingly values digital literacy. Its flexible design accepts future innovations, ensuring the platform remains relevant as educational needs evolve.

Our product is offers a forward-thinking, ethical, and effective approach to transforming education. It enhances learning outcomes, upholds academic integrity, and prepares students for a technology-driven future.

Product Overview

EducateAI is designed to be the solution that addresses the needs of students, educators, and institutions. Its functionality centers around guided problem-solving. Instead of giving direct answers, the AI encourages students to think critically and work through problems with adaptive assistance that decreases as their skills develop. Real-time feedback helps students understand their mistakes and learn more effectively without replacing instructor evaluation. To support academic integrity, the product includes plagiarism detection and tools to identify AI-generated content.

EducateAI integrates smoothly into popular Learning Management Systems such as Canvas and Blackboard, making it accessible without requiring big changes to existing educational infrastructure. Unlike other AI tools such as Khanmigo or Socratic, EducateAI does not only provide answers or act as a passive tutor, it is designed to empower students. By combining technology with educational practices, EducateAI offers a unique value proposition that addresses both current challenges and future demands.

Our product is guided by six key design principles:

- The platform supports student learning by promoting guided problem-solving and adaptive learning experiences, helping students grow without fostering dependency.
- It enhances teaching efficiency while upholding academic integrity through real-time feedback and advanced plagiarism detection tools.
- Ethical and responsible AI use is ensured through transparent AI interactions and clearly defined usage guidelines.
- Student data privacy and security are prioritized, with full compliance with FERPA and GDPR and customizable privacy settings.
- The product is built for accessibility, offering scalable solutions and compatibility with a range of Learning Management Systems—even in resource-limited institutions.
- AI is integrated as a complement to traditional teaching methods, reinforcing rather than replacing human instruction.

The development plan includes developing partnerships with AI developers, educational experts, and data security specialists. Pilot programs will be launched in collaboration with schools and LMS providers to perfect the product and ensure its effectiveness in real-world educational settings. To further encourage adoption, EducateAI will work with educational institutions to develop training programs that help educators integrate the AI tool into their teaching practices effectively.

EducateAI represents a timely and necessary advancement in educational technology. The platform addresses pressing challenges while preparing students for a future shaped by AI. Its balanced approach combines technological innovation with educational integrity, offering an effective solution for institutions of all sizes. As global demand for AI-powered learning solutions continues to grow, EducateAI stands ready to lead the way and transform education.

Strategic Advice:

EducateAI will help schools avoid the problems associated with static business models. As we saw in our class readings in the Harvard Business School case *Why Bad Things Happen to Good Companies*, many once-dominant firms fell not because they lacked talent or resources, but because they failed to adapt to changing markets. Companies like IBM, Sears, and General Motors succeeded during their early years by fulfilling customer needs and leveraging innovative business designs. However, as the case describes, their success eventually disappeared. These organizations failed to notice when customer preferences shifted, and became resistant to innovation. EducateAI

will eliminate this problems and make sure educational institutions don't fall for the same mistakes.

The theme of the Harvard case is value migration. Companies attract value when they meet market demands and create innovative solutions, but over time, as customer needs evolve and new competitors emerge, value migrates away from outdated business models. For IBM, the failure to transition from mainframe computers to the emerging personal computer market led to a massive loss of market share and shareholder value. Sears and General Motors similarly failed as more innovative competitors introduced business models better suited to new customer expectations. Their leadership focused on refining existing operations rather than preparing for structural shifts in their industries. As a result, they were unable to prevent the outward migration of value.

EducateAI must avoid this scenario by continuously adapting to shifts in educational needs. Like IBM in its early years, EducateAI has the advantage of being designed from the ground up for today's market realities. Its foundation in adaptive learning, privacy compliance, and ethical AI use places it as one of the most important platforms of educational technology trends. However, as *Why Bad Things Happen to Good Companies* warns, success can lead to tunnel vision. The same customer-focused innovation that drives growth can, over time, create organizational calcification and a resistance to change.

A crucial step is to consistently track where value is migrating in the education market. As seen in the case, companies in the dynamic competitive equilibrium phase often focus on incremental improvements and tactical competition. EducateAI should avoid this trap by maintaining a strategic perspective, monitoring evolving student learning preferences, educator needs, and institutional budget constraints. This means not only refining existing features like guided problem-solving and adaptive difficulty but also exploring new functionalities, such as AI-driven career guidance, enhanced accessibility features, and integrations with emerging technologies like virtual and augmented reality.

Leadership will play a pivotal role in ensuring that EducateAI does not fall into the complacency observed in the case study's failing firms. As described in the case, effective leadership mobilizes resources and initiates change before a crisis forces the organization's hand. Leaders at EducateAI must foster a culture of proactive experimentation and responsiveness. They should prioritize regular engagement with students, educators, and institutional partners to gather feedback and spot early signs of changing needs.

In my script presentation, I applied these lessons by recommending that educational institutions like Illinois Institute of Technology not only adopt AI solutions like EducateAI but also create AI-in-Education teams composed of students, professors, and technology professionals. These teams would explore new tools, test innovations,

and serve as an early warning system for shifts in educational technology needs. This approach mirrors the successful strategies of companies that, according to the case study, embraced change early rather than reacting to it too late. Establishing such teams within EducateAI's partner institutions will generate new ideas and ensure that the product remains at the cutting edge.

Moreover, the case emphasizes the danger of leadership focusing inward and upward—toward internal politics and historical successes—rather than outward and forward, toward customers and future trends. To counteract this tendency, EducateAI's leadership must regularly engage not only with high-level institutional decision-makers but also with end-users: students and educators. By listening to the challenges they face and the aspirations they hold, EducateAI can remain grounded in real-world needs rather than internal assumptions. This commitment to customer engagement will help prevent the kind of market myopia described in the case, where companies lost touch with evolving customer priorities until it was too late.

The case also warns against over-investment in legacy systems and outdated business models. EducateAI must be cautious not to become overly reliant on existing integrations and product features. As technology and educational practices evolve, the company should be prepared to phase out or radically redesign older components of the platform. This flexibility will require disciplined resource allocation.

Finally, the case maximises the importance of creating a corporate culture that values truth and adaptability over loyalty and tradition. As firms like GM and Sears calcified, they developed cultures that rewarded loyalty over competence. EducateAI's leadership will cultivate an environment where employees at all levels feel empowered.

In conclusion, the lessons from Why Bad Things Happen to Good Companies provide a clear example for EducateAI to follow as it grows. By staying on top of value migration, fostering proactive leadership, engaging continuously with customers and avoiding over-investment in legacy systems, EducateAI can avoid the fate of once-great companies that failed to adapt.

Additional Research:

For EducateAI to succeed as a leader in ethical, AI-driven educational technology, understanding market demand and the broader business landscape is crucial. My research expands upon the initial market size analysis presented in the concept development stage and explores four additional areas of research: adoption barriers and drivers, competitor landscape, regulatory and privacy trends, and long-term revenue scalability. Together, these analyses provide a data-driven foundation for the product's viability and long-term strategic growth.

Expanded Market Size Analysis

The initial research estimated a total addressable market of approximately \$400 million annually, based on 4,000 accredited U.S. higher education institutions, each potentially investing \$10,000 per year in AI tools. Assuming 40 percent readiness for adoption, this would equate to a serviceable available market of \$160 million annually. Even a modest 5 percent market capture would yield \$8 million in annual recurring revenue within three to five years.

To validate and extend these projections, it is important to factor in recent market growth trends and technology adoption rates. According to MarketsandMarkets, the global AI in education market is projected to grow at a compound annual growth rate of 30 percent from 2023 to 2028. Applying this growth rate to the U.S. SAM indicates that the potential market size could expand from \$160 million to nearly \$600 million by 2028.

Assuming 30% CAGR based on MarketsandMarkets projections.

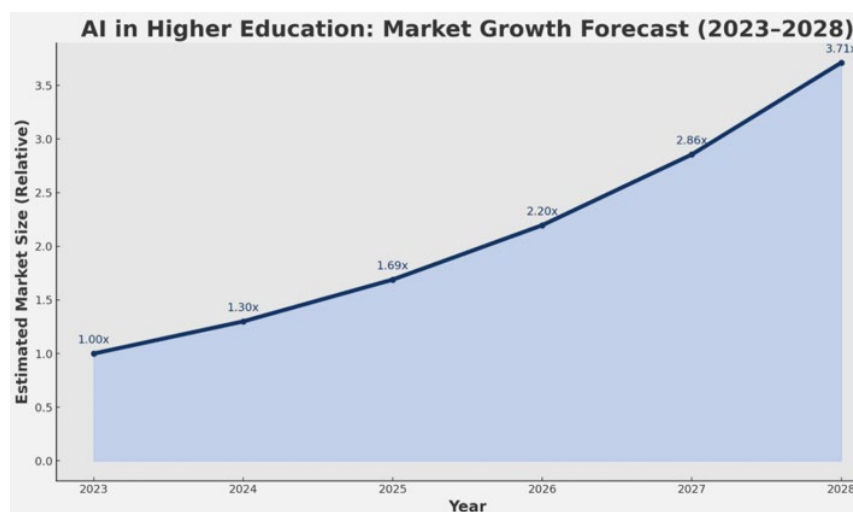


Figure 1: Projected Growth of EducateAI's U.S. Market Opportunity (2023–2028)

By offering a product that is built on infrastructure and designed for remote integration, EducateAI is well-positioned to capitalize on this rapid expansion. It also reinforces the need for scalable, ethical AI solutions that educational institutions can adopt easily as cloud adoption increases globally.

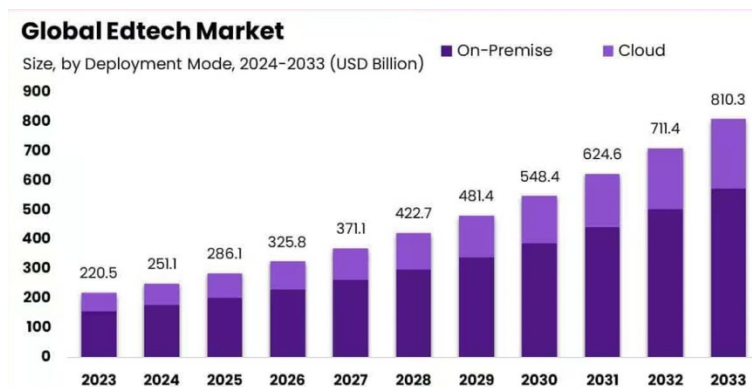


Figure 2: Projected Growth of the Global EdTech Market by Deployment Mode (2023–2033)

Figure 2 highlights the projected growth of the global educational technology (EdTech) market from 2023 through 2033, segmented by deployment mode: on-premise and cloud-based solutions. The market is expected to grow from \$220.5 billion in 2023 to over \$810 billion by 2033..

This trend demonstrates an accelerating shift in education towards digital platforms, particularly cloud-based tools, which are gaining market share due to their flexibility, scalability, and lower infrastructure costs. This growth supports the strategic timing for launching EducateAI with the market’s direction..

Adoption Barriers and Drivers

While the market potential is significant, the rate of adoption will depend on several drivers and barriers. Drivers include the growing demand for personalized learning, increased teacher workload leading to interest in automation, and rising expectations for data-driven educational decision-making. Additionally, the increasing acceptance of blended learning environments provides fertile ground for EducateAI’s LMS-integrated approach.

However, barriers include institutional resistance to change, data privacy concerns, and budget constraints, especially among public institutions. However, as demonstrated in the Harvard case study *Why Bad Things Happen to Good Companies*, the failure to innovate and adapt can lead to market irrelevance. This awareness is expected to accelerate AI adoption as institutions seek to remain competitive and responsive to student needs.

Competitor Landscape

EducateAI will enter a competitive landscape that includes both established players and emerging startups. Competitors like Khanmigo and Socratic offer AI-powered tutoring, but they differ significantly from EducateAI in that they often provide direct answers rather than fostering independent critical thinking. These platforms typically

serve K-12 students or casual learners and do not prioritize integration with higher education LMS platforms.

Other competitors, such as Turnitin and GPTZero, focus primarily on plagiarism detection and AI content monitoring. While they share EducateAI's commitment to academic integrity, they do not offer comprehensive learning support tools.

EducateAI's competitive advantage lies in its unique blend of guided problem-solving, adaptive difficulty, real-time feedback, ethical AI use guidelines, and robust privacy protections that comply with FERPA and GDPR regulations.

Regulatory and Privacy Trends

Compliance with privacy regulations is a non-negotiable requirement for educational technology. The Family Educational Rights and Privacy Act (FERPA) governs the privacy of student education records in the United States, while the General Data Protection Regulation (GDPR) applies to any data collected from students residing in the European Union.

The U.S. Department of Education has indicated that data privacy and ethical AI use will be central to future education policy. EducateAI's design already anticipates these trends by including opt-out mechanisms for AI tracking, encryption of personal data, and audit trails to ensure transparency.

Moreover, the development of AI ethics frameworks by organizations such as UNESCO and EDUCAUSE suggests that ethical AI use will become a standard expectation rather than a competitive differentiator. EducateAI's early commitment to these principles will help establish trust with institutions and position the product as a leader in responsible innovation.

Long-Term Revenue Scalability

First, expansion beyond U.S. higher education into K-12 schools, and international markets could exponentially increase the addressable market. The global AI-in-education market is expected to reach \$20 billion by 2028, offering opportunity for vertical and geographic growth. Potential partnerships with LMS providers and educational content publishers can create new revenue streams.

EducateAI can introduce additional features such as predictive analytics for student performance, AI-driven course recommendations, and integration with career services platforms. These enhancements will not only increase product value but also justify premium pricing over time.

The expanded market analysis and additional research areas confirm that EducateAI operates within a rapidly growing, strategically promising sector. By understanding and addressing adoption barriers, differentiating itself within the competitive landscape, and planning for long-term scalability, EducateAI is well positioned for sustained future success. As institutions increasingly recognize the need to innovate, the demand for solutions like EducateAI will only continue to grow.

The data-driven insights and strategic foresight presented in this research will guide EducateAI's development, marketing, and partnership strategies, ensuring that it anticipates the educational challenges and opportunities of the future.

The following pages contain an artifact sample for **Goal 3 Valuation**.

Reflection Report: Financial Evaluation of the Hertz LBO and IPO

Course: BUS 455 – Corporate Finance

[REDACTED]

Instructor: Xinze Qian

Date: May 10, 2025

1. Financial Decision Rules

To evaluate whether Alec Berg should invest in the Hertz IPO, we apply three core valuation tools: weighted average cost of capital (WACC), net present value (NPV), and payback period. These tools help determine whether the opportunity presents a favorable risk-adjusted return.

According to Exhibit 11, Hertz's post-LBO capital structure includes \$6,426.1 million in non-fleet corporate debt and \$4,809.3 million in equity (320.62 million shares at \$15 per share), for a total capital base of \$11,235.4 million. The debt-to-equity ratio is 1.34. We assume a cost of equity of 11% (using CAPM) and a cost of debt of 7%, given sub-investment-grade credit quality. With a corporate tax rate of 35%, the WACC is:

$$\text{WACC} = (0.428 \times 11\%) + (0.572 \times 7\% \times (1 - 0.35)) = 4.71\% + 2.60\% = 7.31\%$$

Using RAC Adjusted EBITDA of \$570.3 million and HERC Gross EBITDA of \$750 million from Exhibit 11, total EBITDA equals \$1,320.3 million. Assuming 60% of EBITDA converts to unlevered free cash flow (FCF), we estimate year 1 FCF at \$792.2 million.

Assuming 5% EBITDA growth and a 3% terminal growth rate, the terminal value at year 5 is:

$$TV = 962.9 \times 1.03 / (0.0731 - 0.03) = 991.8 / 0.0431 \approx \$22,999 \text{ million}$$

Discounting all cash flows to present using a 7.31% WACC yields a total present value of \$19,657 million. The initial investment (enterprise value) equals \$13,955.5 million in debt plus \$4,809.3 million in equity = \$18,764.8 million.

$$NPV = \$19,657M - \$18,764.8M = \$892.3 \text{ million}$$

The investment is financially attractive. Berg's payback period is approximately 1.64 years:

$$\text{Payback Time} = 1 + (1,323 - 792.2) / 831.8 \approx 1.64 \text{ years}$$

This aligns well with private equity horizons of 3–5 years and supports investment.

2. After-Tax Cash Flow Analysis

An after-tax cash flow analysis demonstrates how debt benefits valuation through the interest tax shield. With \$6.43B in corporate debt and a 7% interest rate, Hertz's annual tax shield equals:

$$\text{Tax Shield} = 6,426.1 \times 0.07 \times 0.35 = \$157.5 \text{ million}$$

This reduces the effective cost of debt from 7% to 4.55%. Using this adjustment, WACC falls to 7.31%, enhancing DCF valuation.

The present value of a perpetual tax shield, per Brealey et al. (2020), is:

$$\text{PV of Tax Shield} = \text{Debt} \times \text{Tax Rate} = 6,426.1 \times 0.35 = \$2,249.1 \text{ million}$$

Hertz's capital structure created substantial value via this mechanism. The estimate assumes debt levels remain constant, a reasonable assumption for LBOs.

We convert EBITDA to unlevered FCF using McKinsey's 60% rule (McKinsey & Company, 2020). Total EBITDA from Exhibit 11 is \$1,320.3 million:

Unlevered FCF = $1,320.3 \times 0.60 = \$792.2$ million

If tax rates decline (e.g., 25%): Tax Shield = \$112.5 million/year

If tax rates rise (e.g., 40%): Tax Shield = \$179.9 million/year

Tax policy materially affects valuation. Future tax reform would influence investment outcomes for Alec Berg, reducing the attractiveness of highly leveraged deals like Hertz's if rates fall.

3. Debt Financing Questions

The Hertz LBO, closed in December 2005, involved \$2.3 billion in sponsor equity and \$12.7 billion in debt (Chaplinsky et al., 2009). Exhibit 11 shows the breakdown: \$6.4B in corporate loans and \$7.5B in fleet ABS.

ABS financing allowed Hertz to obtain AAA-rated debt secured by fleet cash flows, isolating fleet risk. Fleet assets were held in an SPV, which allowed Hertz to lease the vehicles back and preserve tax benefits.

Post-LBO, Hertz's debt-to-EBITDA ratio = $\$13.96\text{B} / \$1.32\text{B} = 10.6\times$ — a highly leveraged position. Exhibit 10 reveals that of 222 LBOs Moody's reviewed, 45% involved special dividends, and 58 were downgraded post-deal.

In June 2006, sponsors added \$1 billion in debt to pay themselves a dividend. This occurred before appointing a new CEO. Such a dividend recap significantly weakened Hertz's capital base.

Lenders in 2006 offered PIK-toggle notes and covenant-lite loans. Flexed pricing helped underwriters place riskier tranches. These innovations enabled the LBO but left the post-deal company financially fragile.

Investors evaluating such transactions must assess not only financial projections but also the sustainability of capital structure under real-world stress.

4. Equity Financing Questions

Hertz's sponsors pursued an IPO over a private sale to retain control and liquidity. At IPO, they retained 71.9% of shares, maintaining board authority (Hertz S-1, 2006).

The IPO raised \$1.425 billion. However, none of the proceeds were retained by the company. They were used to repay a loan facility tied to a \$1 billion dividend paid to the sponsors.

This raised governance concerns. Investors saw little reinvestment in the business.

BusinessWeek (2007) described such "strip-and-flip" strategies as symptomatic of short-term sponsor motives.

IPO pricing was reduced from \$16–\$18 to \$15 due to tepid demand. Risk factors included leverage, sponsor overhang, and a lack of operational transformation.

Equity value = $320.6\text{M} \times \$15 = \4.809B

Debt = \$13.955B (Exhibit 11), Cash = \$441M

Enterprise Value = $4.809 + 13.955 - 0.441 = \18.323B

Equity/EV = 26.2%

This capital structure signaled high risk for public investors, with sponsors retaining significant upside and limited accountability.

5. Comparative Analysis Questions

Debt created tax efficiency and leveraged returns, but equity financing would have reduced default risk and improved strategic flexibility.

If Hertz had retained earnings or issued more equity, it could have avoided the 10.6× debt/EBITDA ratio and improved financial sustainability.

A dual-track strategy—preparing for an IPO and seeking private bids—would have created competitive tension and possibly increased the exit multiple (PwC, 2022).

Alternatively, if Hertz had remained private longer, sponsors could have reduced debt through FCF and improved IPO story with operational improvements and stronger leadership (CEO appointed only weeks before IPO).

According to Cao and Lerner (2009), many reverse LBOs underperform due to overleverage and premature public offerings.

Compared to Enterprise (privately owned) and Avis (lower leverage), Hertz's fixed obligations constrained agility in pricing and fleet investments—critical in the cyclical car rental industry.

Thus, while the LBO created near-term sponsor value, it compromised post-IPO flexibility. Investors like Alec Berg must weigh the short-term NPV against long-term solvency and governance risks.

Conclusion:

Despite the appearance of strong returns, the Hertz IPO is not a sound investment. Its highly leveraged capital structure, use of aggressive financial engineering, and the fact that IPO proceeds benefited the sponsors—not the company—signal misaligned incentives and elevated risk. For a prudent, long-term investor like Alec Berg, the downside risks far outweigh the potential rewards.

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